www.tntextbooks.in

activity recorded by EEG may have synchronised or desynchronised waves. It has four frequency waves/ rhythms namely alpha, beta, delta and theta waves.

## Clinical significance of EEG

1. EEG provides a means to study the functioning of the brain and its coordination with other parts of the body.
2. It is useful in diagnosis of neurological and sleep disorders
3. EEG has proved to be a useful diagnostic tool in cases of serious head injuries, brain tumours and cerebral infections.
4. It also helps to find the diseases like epilepsy and various degenerative disease of the nervous system.
5. EEG is useful in assessing patients with suspected brain death.

## X-rays

Radiography is the use of X-rays to visualize the internal structures of a patient (Figure12.6). X-Rays are a form of electromagnetic radiation, produced by an X-ray tube. The X-rays are passed through the body and captured behind the patient by a detector; film sensitive to x-rays or a digital detector. There is variation in absorption of the X-rays by different tissues within the body. Dense bones absorb more radiation, while soft tissue allows more rays to pass through. This variation produces contrast image within the image to give a 2D representation of all the structures. It is invasive. It is also available as a portable X ray unit. It is less costly when compared to the other imaging units like MRI or CT.

**Figure 12.6.** X Ray machine

'X rays were discovered by German physicist Sir Wilhelm Conard Roentgen in 1895, while he was studying on high voltage discharge phenomenon in a Crooker's tube. He noticed fluorescence of a barium platino cyanide screen lying several feet away in the same room.

## Clinical significance

1. X- ray imaging is used for diagnosing the disease of the heart, lungs and fractures of bones and joints
2. It is also used to visualise hollow organs and blood vessels by filling them with certain chemical formulations containing barium and iodine
3. Dental radiography is used in diagnosis of oral problems